

HelpDesk Ticketing System Documentation

Overview

The HelpDesk Ticketing System is a web-based support management platform designed to manage support requests, user concerns, and issue tracking within an organization. The system allows administrators and users to create, monitor, and manage support tickets efficiently.

This documentation provides a complete guide for installation, configuration, setup, usage, and troubleshooting of the system.

System Information

System Name

HelpDesk Ticketing System

System Type

Web-Based Ticketing and Support Management System

Developed Using

- Node.js
- Express.js
- Microsoft SQL Server
- HTML/CSS/JavaScript
- Bootstrap

AI-Assisted Development

This project was developed with the assistance of AI tools including:

- Claude AI
 - OpenAI Codex
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Features

User Features

- User login authentication
- Submit support tickets
- View ticket status
- Update existing tickets
- Dashboard overview
- Ticket history tracking

Admin Features

- Manage users
 - Manage tickets
 - Assign ticket priorities
 - Monitor ticket progress
 - View reports and logs
 - Manage departments and categories
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System Requirements

Hardware Requirements

Minimum

- Intel Core i3 Processor
- 8GB RAM
- 256GB SSD/HDD
- Stable Internet or LAN Connection

Recommended

- Intel Core i5/i7 or Ryzen 5/7
 - 16GB RAM
 - SSD Storage
 - Gigabit LAN Connection
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Software Requirements

Required Applications

- Windows 10 or Windows 11
 - Visual Studio Code
 - Node.js LTS Version
 - Microsoft SQL Server 2022 Express
 - SQL Server Management Studio (SSMS)
 - Git for Windows
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Installation Guide

Step 1 – Install Microsoft SQL Server

1. Download Microsoft SQL Server Express.
 2. Run the installer.
 3. Choose:
 4. Basic Installation
 5. Wait for installation to finish.
 6. Install SQL Server Management Studio (SSMS).
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Step 2 – Open SQL Server Management Studio

Login Settings

Server Name

localhost\SQLEXPRESS

Authentication

Windows Authentication

Important

Enable:

- Trust server certificate

Click:

- Connect
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Step 3 – Create Database

1. Expand:
 2. Databases
 3. Right click:
 4. New Database
 5. Database Name:
 6. TicketingDB
 7. Click OK.
-

Step 4 – Import Database Script

1. Open the SQL file:
2. database_setup.sql
3. Select database:
4. TicketingDB
5. Click Execute.

If successful, the system will automatically create:

- Tables
- Default users

- Roles
 - Ticket structures
 - Initial data
-

Default Login Credentials

Super Administrator

Username

superadmin

Password

Admin@123

Important: Change the default password after first login.

Step 5 – Install Node.js

1. Download Node.js LTS version.
 2. Install Node.js.
 3. Enable:
 4. Add to PATH
 5. Finish installation.
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Step 6 – Open the Project in VS Code

1. Open Visual Studio Code.
 2. Click:
 3. File → Open Folder
 4. Select the ticketing-system folder.
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Step 7 – Install Project Dependencies

Open terminal in VS Code:

```
npm install
```

Step 8 – Configure Environment Variables

Create or edit the .env file.

Sample Configuration

```
DB_SERVER=127.0.0.1
DB_PORT=1433
DB_NAME=TicketingDB
DB_USER=sa
DB_PASSWORD=YourPassword123
DB_ENCRYPT=false
DB_TRUST_SERVER_CERT=true
PORT=3000
NODE_ENV=development
```

Step 9 – Enable TCP/IP in SQL Server

1. Open SQL Server Configuration Manager.
2. Go to:
3. SQL Server Network Configuration
4. Protocols for SQLEXPRESS
5. Enable:
6. TCP/IP
7. Set Port:
8. 1433

9. Restart SQL Server.

Step 10 – Run the System

Open VS Code terminal:

```
npm run dev
```

If the system does not use dev mode:

```
npm start
```

Accessing the System

Open browser:

```
http://localhost:3000
```

Git Setup

Install Git

1. Install Git for Windows.
2. Recommended settings:
3. Checkout Windows-style, commit Unix-style line endings
4. Use MinTTY

Verify Git Installation

```
git --version
```

Using GitHub

Initialize Repository

```
git init
```

Add Files

```
git add .
```

Commit Changes

```
git commit -m "Initial commit"
```

Push to GitHub

```
git remote add origin YOUR_GITHUB_REPOSITORY  
  
git push -u origin main
```

VS Code Extensions Recommendation

Recommended Extensions

- OpenAI Codex
- Prettier
- ESLint
- SQLTools
- GitLens

Common Errors and Solutions

SQL Connection Error

Problem

Failed to connect to localhost:1433

Solution

- Enable TCP/IP
 - Restart SQL Server
 - Check DB_PORT
 - Verify SQL credentials
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Invalid Credentials

Problem

Cannot login using default account.

Solution

- Verify imported database
 - Reset password hash
 - Ensure user exists in database
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npm Command Not Found

Solution

Install Node.js and restart terminal.

Recommended Folder Structure

Recommended project location:

D:\Projects\ticketing-system

Avoid storing development projects inside OneDrive folders to prevent path and sync issues.

Security Recommendations

- Change default passwords
- Use strong SQL passwords
- Restrict database access
- Enable firewall rules
- Backup the database regularly
- Use HTTPS in production deployment

Backup Recommendation

Database Backup

Use SQL Server backup feature regularly.

Project Backup

Backup:

- Source code
- .env configuration
- Database backups
- Uploaded files

Future Improvements

Potential future improvements:

- Email notifications
 - SMS integration
 - Real-time chat support
 - File attachment previews
 - Analytics dashboard
 - Mobile responsive optimization
 - Multi-branch support
 - Cloud deployment
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Deployment Suggestions

Possible hosting options:

- Render
- Vercel
- Railway
- VPS Hosting
- Microsoft Azure

Conclusion

The HelpDesk Ticketing System provides an efficient solution for managing technical support requests and organizational issue tracking. With proper configuration and deployment, the system can support small to medium-sized businesses and organizations.

This documentation serves as a complete installation and operational guide for developers, administrators, and future maintainers of the project.

Credits

System Documentation Prepared By: Mark John Rey Espiritu

AI-Assisted Development Using:

- Claude AI
- OpenAI Codex

Special Credits: Dezu